

SECTION A – 5 MARK QUESTIONS

Q1. Differentiate between an analog image and a digital image. Explain how a digital image is represented mathematically.

Analog Image

An analog image is a continuous representation of a real-world scene. Its spatial coordinates and intensity values can vary continuously.

Examples: Photograph on film, traditional X-ray film, painting.

Digital Image

A digital image is an image represented using a finite number of pixels. Both spatial coordinates and intensity values are discrete.

Examples: JPEG photograph, PNG image, digital X-ray.

Difference

Analog Image	Digital Image
Continuous in nature	Discrete in nature
Infinite number of possible intensity values	Finite intensity values
Difficult to process using computers	Easily processed by computers
Can degrade during copying	Can be copied with little/no quality loss
Stored on film or analog media	Stored as binary data
Processing is comparatively difficult	Processing is easier

Mathematical Representation

A digital image can be represented as a two-dimensional function:

$$f(x,y)$$

where:

- x = horizontal coordinate
- y = vertical coordinate
- $f(x,y)$ = intensity value at coordinate (x,y)

For a digital image:

$$x=0,1,2,\dots,M-1 \quad y=0,1,2,\dots,N-1$$

Therefore, an image can be represented as an $M \times N$ matrix:

$$f(x,y) = \begin{bmatrix} f(0,0) & f(0,1) & \dots & f(0,N-1) \\ f(1,0) & f(1,1) & \dots & f(1,N-1) \\ \vdots & \vdots & \ddots & \vdots \\ f(M-1,0) & f(M-1,1) & \dots & f(M-1,N-1) \end{bmatrix}$$

Each element represents one **pixel**.

Q2. Explain the different components involved in a Digital Image Processing system.

A Digital Image Processing (DIP) system consists of hardware and software components used to acquire, process, store and display images.

Main Components

1. **Image Acquisition**
 - Captures an image using cameras, scanners or sensors.
 - Converts the physical image into digital data.
2. **Specialized Image Processing Hardware**
 - Performs high-speed image processing.
 - Examples include GPUs, image-processing boards and dedicated processors.
3. **Computer**
 - Controls the image-processing operations.
 - Executes processing algorithms.
4. **Image Processing Software**
 - Contains algorithms for enhancement, restoration, segmentation, etc.
 - Examples: MATLAB, OpenCV and Python libraries.
5. **Mass Storage**
 - Stores images and processed results.
 - Examples: SSD, HDD, cloud storage.
6. **Image Display**
 - Displays the processed image.
 - Examples: monitor, projector and medical display.
7. **Hardcopy Devices**
 - Produce permanent copies.
 - Examples: printers and plotters.
8. **Communication Network**
 - Transfers images between systems.

- Important in applications such as telemedicine and satellite imaging.

Basic Flow

Image → Acquisition → Processing → Storage → Display/Output

Q3. Classify digital images and explain the major types of images with suitable examples.

Digital images can be classified according to the number of intensity or color values used.

1. Binary Image

A binary image contains only two intensity values:

0 and 1

Usually:

- 0 = black
- 1 = white

Example: Scanned document containing only black and white text.

2. Grayscale Image

A grayscale image contains different shades between black and white.

For an 8-bit grayscale image:

0–255

are possible intensity levels.

- 0 = black
- 255 = white

Example: X-ray image.

3. Color Image

A color image contains multiple color channels. RGB is commonly used.

An 8-bit RGB image has:

- Red: 0–255

- Green: 0–255
- Blue: 0–255

Therefore, it can represent approximately:

$$256^3 = 16,777,216$$

colors.

Example: Digital camera photograph.

4. Multispectral Image

Contains several spectral bands beyond normal RGB.

Example: Satellite images used for agriculture and land analysis.

5. Hyperspectral Image

Contains a very large number of narrow spectral bands.

Example: Remote sensing and mineral identification.

Q4. Explain any five commonly used image file formats such as BMP, JPEG, PNG, GIF and TIFF.

1. BMP – Bitmap

- Developed for storing bitmap images.
- Usually uses little or no compression.
- Provides good image quality.
- File size is generally large.

Application: Windows graphics and simple image storage.

2. JPEG – Joint Photographic Experts Group

- Uses lossy compression.
- Produces small file sizes.
- Suitable for photographs.
- Some image quality is lost during compression.

Application: Digital photographs and websites.

3. PNG – Portable Network Graphics

- Uses lossless compression.
- Supports transparency.
- Provides good image quality.
- Suitable for graphics and screenshots.

Application: Web graphics, logos and transparent images.

4. GIF – Graphics Interchange Format

- Supports up to 256 colors per frame.
- Uses lossless compression.
- Supports simple animation.
- Suitable for simple graphics.

Application: Web animations and small graphics.

5. TIFF – Tagged Image File Format

- Supports high-quality images.
- Can use lossless compression.
- Supports high color depth.
- Commonly used for professional and archival purposes.

Application: Medical imaging, scanning and publishing.

Comparison

Format	Compression	Main Feature
BMP	Usually uncompressed	Simple, large files
JPEG	Lossy	Small size, photographs
PNG	Lossless	Transparency
GIF	Lossless	Animation, 256 colors
TIFF	Lossless/lossy options	High-quality archival images

Q5. Explain the fundamental steps involved in Digital Image Processing.

The major steps are:

1. Image Acquisition

The image is captured using a sensor or camera and converted into digital form.

2. Image Enhancement

Improves the visual appearance of an image.

Examples:

- Contrast enhancement
- Brightness adjustment
- Sharpening

3. Image Restoration

Attempts to recover an image degraded by noise or blur.

4. Color Image Processing

Processes images containing color information.

5. Wavelets and Multiresolution Processing

Represents an image at different resolutions or frequency levels.

6. Image Compression

Reduces the amount of storage required.

7. Morphological Processing

Processes the shape and structure of objects.

8. Image Segmentation

Divides an image into meaningful regions or objects.

9. Representation and Description

Represents detected objects and extracts their features.

10. Object Recognition

Identifies objects in an image.

11. Knowledge Base

Contains information that guides the processing system.

General Flow

Acquisition → Enhancement/Restoration → Segmentation → Representation → Description → Recognition → Output

Q6. Explain the basic elements of human visual perception relevant to Digital Image Processing.

Human visual perception describes how the eye and brain perceive images.

1. Human Eye

The eye acts as an image-sensing system.

Important parts include:

- Cornea
- Lens
- Iris
- Retina

2. Retina

The retina contains photoreceptors:

- **Rods** – sensitive to low light.
- **Cones** – responsible for color and detailed vision.

3. Brightness Adaptation

The human eye can adjust to different levels of illumination.

For example, the eye can adapt from a dark room to bright sunlight.

4. Contrast Sensitivity

Humans perceive differences between neighboring intensity levels.

An object with high contrast is usually easier to see.

5. Color Perception

Cones are responsible for color perception. Different cone responses help humans distinguish different colors.

6. Spatial Resolution

The eye can distinguish small details depending on viewing conditions and contrast.

Importance in DIP

Knowledge of human perception helps in:

- Image enhancement
 - Contrast adjustment
 - Image compression
 - Display design
 - Medical image analysis
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Q7. Explain image sensing and acquisition. Describe how an image is converted into a digital form.

Image Sensing

Image sensing is the process of detecting light or other electromagnetic energy from a scene.

Sensors include:

- CCD sensors
- CMOS sensors
- X-ray detectors
- MRI sensors

Image Acquisition

Image acquisition converts the sensed physical information into a digital image.

Process

Real-world scene → Sensor → Analog signal → Sampling → Quantization → Digital image

Step 1: Illumination

The object is illuminated or produces energy itself.

Step 2: Sensing

A sensor detects the energy reflected or emitted by the object.

Step 3: Analog Signal

The sensor produces a continuous electrical signal.

Step 4: Sampling

The continuous spatial coordinates are converted into discrete coordinates.

Step 5: Quantization

The continuous intensity values are converted into discrete intensity levels.

Step 6: Digital Storage

The resulting values are stored as binary data.

Q8. Differentiate between image sampling and image quantization. Explain their effect on spatial and intensity resolution.

Sampling

Sampling converts continuous spatial coordinates into discrete coordinates.

It determines the number of pixels in an image.

Sampling mainly affects spatial resolution.

Quantization

Quantization converts continuous intensity values into a finite number of intensity levels.

For an n -bit image:

$$L = 2^n$$

where L is the number of intensity levels.

Difference

Sampling	Quantization
Deals with spatial coordinates	Deals with intensity values
Determines number of pixels	Determines number of intensity levels
Affects spatial resolution	Affects intensity resolution
Low sampling can cause loss of detail	Low quantization can cause false contours

Example

A 1000×1000 image has more spatial samples than a 250×250 image.

An 8-bit image has:

$$2^8 = 256$$

intensity levels.

A 4-bit image has:

$$2^4 = 16$$

intensity levels.

Q9. Explain the basic relationships between pixels: neighborhood, adjacency and connectivity.

For a pixel $p(x,y)$, its relationships with surrounding pixels are important.

1. 4-Neighborhood

The four horizontal and vertical neighbors are:

$$N_4(p) = \{(x-1, y), (x+1, y), (x, y-1), (x, y+1)\}$$

2. 8-Neighborhood

Includes the four diagonal pixels as well:

$$N_8(p) = N_4(p) + \text{four diagonal neighbors}$$

3. Adjacency

Two pixels are adjacent if they are neighbors and satisfy a specified condition, usually belonging to the same set of pixel values.

4. 4-Adjacency

Two pixels are 4-adjacent if one is in the 4-neighborhood of the other.

5. 8-Adjacency

Two pixels are 8-adjacent if one belongs to the 8-neighborhood of the other.

6. Connectivity

Connectivity determines whether pixels belong to the same connected object.

It is used in:

- Object detection
- Region extraction
- Image segmentation
- Boundary detection

Q10. Compare RGB, CMY(K), HSI and YCbCr color models and mention one suitable application of each.

Model	Components	Main Use
RGB	Red, Green, Blue	Displays
CMY(K)	Cyan, Magenta, Yellow, Black	Printing
HSI	Hue, Saturation, Intensity	Color segmentation
YCbCr	Y, Cb, Cr	Video/image compression

RGB

An additive color model.

$RGB=(R,G,B)$

Used in:

- Computer monitors
- TVs
- Digital cameras

CMY(K)

A subtractive color model.

- C = Cyan
- M = Magenta
- Y = Yellow
- K = Black

Used mainly in printing.

HSI

- H = Hue
- S = Saturation
- I = Intensity

It separates color information from brightness.

Used in color-based image processing.

YCbCr

- Y = Luminance
- Cb = Blue chrominance
- Cr = Red chrominance

Used in digital video and image compression.

SECTION B – 10 MARK QUESTIONS

**Q1. Explain analog and digital images in detail.
Describe digital image representation, image**

classification, types of images and commonly used image file formats with suitable examples.

Analog Image

An analog image is continuous in both spatial coordinates and intensity.

For example, an image formed on photographic film contains continuously varying brightness.

Digital Image

A digital image is obtained by sampling and quantizing an analog image.

It consists of a finite number of pixels.

Mathematically:

$$f(x,y)$$

represents the intensity at position (x,y) .

A digital image is represented as:

$$M \times N \text{ times } N$$

pixels.

Digital Image Representation

For an 8-bit grayscale image:

$$0 \leq f(x,y) \leq 255$$

For an RGB image, each pixel contains three values:

$$(R,G,B)$$

Classification of Images

1. Binary images
2. Grayscale images
3. Color images
4. Multispectral images
5. Hyperspectral images

Common Formats

BMP: Simple bitmap format, usually large.

JPEG: Lossy compression, excellent for photographs.

PNG: Lossless compression and transparency.

GIF: Limited colors and animation.

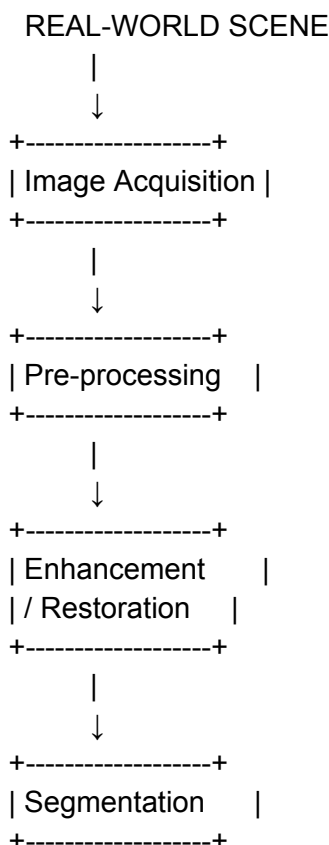
TIFF: High-quality format suitable for professional and archival applications.

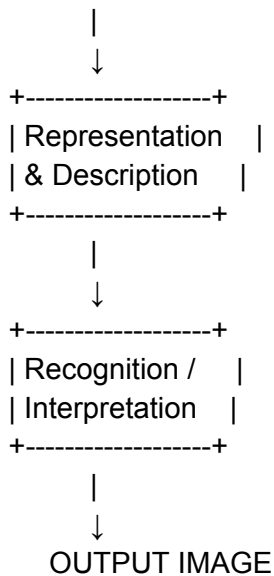
Conclusion

Digital images are preferred for computer-based processing because they can be easily stored, transmitted, enhanced, analyzed and compressed.

Q2. Draw and explain the complete Digital Image Processing system. Discuss the fundamental processing steps and major hardware/software components involved from image acquisition to output.

Digital Image Processing System





1. Image Acquisition

A camera or sensor captures the scene.

It includes:

- Sensing
- Sampling
- Quantization

2. Pre-processing

Removes unwanted effects and prepares the image.

Examples:

- Noise removal
- Resizing
- Filtering

3. Enhancement

Improves visual appearance.

Examples:

- Contrast enhancement
- Sharpening
- Histogram processing

4. Restoration

Attempts to recover an image affected by blur or noise.

5. Segmentation

Divides an image into meaningful regions.

6. Representation and Description

Objects are represented using features such as:

- Shape
- Area
- Boundary
- Texture

7. Recognition

Identifies objects.

For example, identifying a tumor in a medical image.

Hardware Components

- Camera/sensor
- Computer
- GPU
- Image acquisition card
- Storage
- Display
- Printer
- Network devices

Software Components

- Operating system
 - Image-processing software
 - Algorithms
 - Libraries such as OpenCV
 - Programming languages such as Python/C++
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Q3. Explain image sensing, acquisition, sampling and quantization in detail. Discuss spatial resolution, intensity resolution and the effects of changing sampling rate and quantization levels on image quality.

Image Sensing

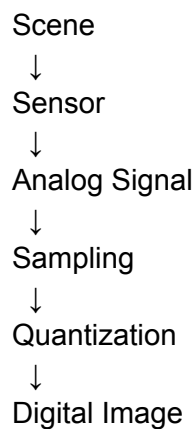
Image sensing is the detection of energy from a scene using a sensor.

Examples:

- CMOS camera sensor
- CCD
- X-ray detector
- MRI sensor

Image Acquisition

The sensor output is converted into digital data.



Sampling

Sampling determines the spatial resolution.

If the image has:

$M \times N$ samples,

then it contains $M \times N$ pixels.

High Sampling Rate

- More pixels

- More spatial detail
- Larger storage requirement

Low Sampling Rate

- Fewer pixels
- Loss of fine details
- Possible aliasing

Quantization

Quantization determines the number of intensity levels.

For b bits:

$$L = 2^b$$

For 8 bits:

$$L = 256$$

For 4 bits:

$$L = 16$$

Spatial Resolution

Spatial resolution refers to the ability to distinguish small objects or details.

Higher resolution generally means more image detail.

Intensity Resolution

Intensity resolution refers to the ability to distinguish small differences in brightness.

Higher bit depth provides more intensity levels.

Effect of Sampling and Quantization

Change	Effect
Increase sampling	Better spatial detail
Decrease sampling	Loss of spatial detail
Increase quantization levels	Smoother intensity transitions
Decrease quantization levels	False contours/banding

Higher sampling + quantization Better quality but more storage

Q4. Explain color image fundamentals in detail. Compare RGB, CMY(K), HSI and YCbCr color models with their components, characteristics and applications.

Color images contain information about the spectral properties of objects.

1. RGB Model

RGB uses three primary additive colors:

- Red
- Green
- Blue

Each color channel can have values from 0–255 in an 8-bit channel.

Applications:

- Monitors
- Cameras
- Smartphones
- Computer graphics

2. CMY(K) Model

CMY is a subtractive color model.

- Cyan
- Magenta
- Yellow

Black is added as K to improve printing efficiency and quality.

Applications:

- Printers
- Publishing
- Printing industry

3. HSI Model

HSI separates color from intensity.

Hue

Represents the dominant color.

Saturation

Represents color purity.

Intensity

Represents brightness.

Applications:

- Color segmentation
- Object detection
- Image analysis

4. YCbCr Model

YCbCr separates brightness from chrominance.

- **Y** = Luminance
- **Cb** = Blue-difference chroma
- **Cr** = Red-difference chroma

Applications:

- Digital television
- Video
- JPEG-related imaging workflows
- Video compression

Comparison

Feature	RGB	CMY(K)	HSI	YCbCr
Type	Additive	Subtractive	Perceptual	Luma/chroma
Components	R,G,B	C,M,Y,K	H,S,I	Y,Cb,Cr
Main Application	Display	Printing	Segmentation	Video
Brightness separated?	No	No	Yes	Yes

Q5. Explain two-dimensional mathematical preliminaries used in Digital Image Processing. Discuss pixel relationships, image geometry and important mathematical tools/operations used for processing digital images.

Two-dimensional mathematical concepts are important for representing and manipulating digital images.

1. Coordinate System

A pixel is represented by:

(x,y)

where x and y represent its position.

For example:

$p=(3,4)$

represents a pixel at coordinate $(3,4)$.

2. Distance Measures

Euclidean Distance

For pixels $p(x_1,y_1)$ and $q(x_2,y_2)$:

$$D_E = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

City-Block Distance

$$D_4 = |x_1 - x_2| + |y_1 - y_2|$$

Chessboard Distance

$$D_8 = \max(|x_1 - x_2|, |y_1 - y_2|)$$

3. Pixel Neighborhood

Pixels can have:

- 4-neighborhood
- 8-neighborhood

4. Translation

Translation moves an image from one position to another.

$$x' = x + t_x \quad y' = y + t_y$$

5. Scaling

Scaling changes image size.

$$x' = s_x x \quad y' = s_y y$$

6. Rotation

Rotation by angle θ :

$$x' = x \cos \theta - y \sin \theta \quad y' = x \sin \theta + y \cos \theta$$

7. Matrix Operations

Images can be represented as matrices. Common operations include:

- Addition
- Subtraction
- Multiplication
- Convolution
- Filtering

8. Connectivity

Connectivity determines whether pixels form a connected object.

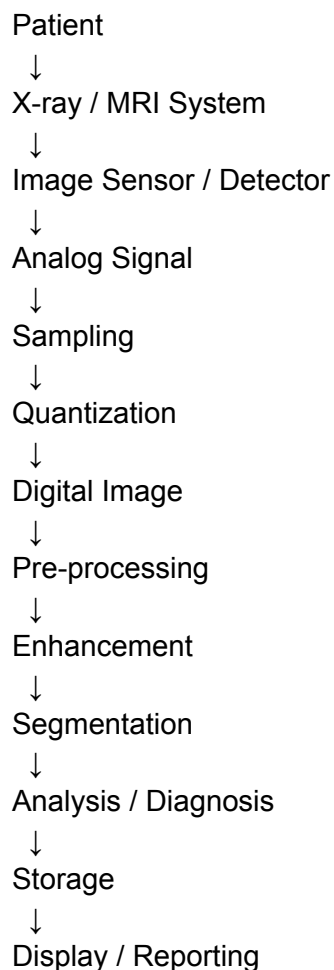
It is useful for:

- Object detection
 - Segmentation
 - Region extraction
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SECTION C – APPLICATION / ANALYTICAL QUESTIONS – 15 MARKS

Q1. A hospital is designing a digital medical-imaging system for processing X-ray/MRI images. Propose a complete Digital Image Processing workflow starting from image sensing and acquisition. Explain sampling, quantization, enhancement, pixel relationships and suitable image representation/file format. Also discuss how spatial and intensity resolution can influence diagnostic image quality.

Complete Medical Image Processing Workflow



1. Image Sensing and Acquisition

For an X-ray system, an X-ray detector captures radiation passing through the patient's body.

For MRI, the system measures signals generated from the interaction of the magnetic field and radiofrequency excitation with tissue.

The detected information is converted into electrical/digital data.

2. Sampling

The continuous spatial information is sampled into pixels.

Higher sampling rates provide more spatial detail.

For medical images, sufficient sampling is important to distinguish small anatomical structures.

3. Quantization

The measured signal intensity is converted into discrete numerical levels.

If 12 bits are used:

$$2^{12} = 4096$$

intensity levels are available.

Higher bit depth allows finer differences in tissue intensity to be represented.

4. Enhancement

Medical images can be enhanced using:

- Contrast enhancement
- Histogram processing
- Noise reduction
- Sharpening

For example, contrast enhancement can make boundaries between tissues easier to observe.

5. Pixel Relationships

Pixel relationships help identify structures.

Important relationships include:

- 4-neighborhood
- 8-neighborhood
- 4-connectivity
- 8-connectivity

These are useful for segmentation and detecting connected anatomical structures.

6. Image Representation

Grayscale representation is commonly suitable for X-ray images because the image primarily represents differences in intensity.

MRI can also be represented using grayscale images, although the underlying acquisition can contain richer numerical information.

7. File Format

For medical imaging, **DICOM** is particularly suitable because it is designed for medical images and can store image data together with relevant metadata.

For general high-quality image processing/archival workflows, TIFF can also be useful, depending on the system requirements.

8. Spatial Resolution

Higher spatial resolution allows smaller anatomical structures to be distinguished.

If resolution is too low:

- Fine structures may disappear.
- Small abnormalities may be difficult to identify.

9. Intensity Resolution

Higher intensity resolution allows subtle differences between tissues to be represented.

If intensity resolution is too low:

- Similar tissues may appear identical.
- Important contrast differences can be lost.

Conclusion

A medical DIP system should maintain sufficient spatial and intensity resolution while using appropriate enhancement, segmentation and storage techniques. The objective is to improve visualization without removing medically important information.

Q2. A digital camera captures a 1920×1080 RGB image with 8 bits for each R, G and B channel.

(a) Calculate the total number of pixels.

Given:

Width=1920 Width=1920 Height=1080 Height=1080

Therefore:

$$\text{Pixels} = 1920 \times 1080 = 2,073,600 \text{ pixels} = \boxed{2,073,600 \text{ pixels}}$$

So the image contains approximately **2.07 megapixels**.

(b) Determine bits per pixel.

There are three channels:

- Red = 8 bits
- Green = 8 bits
- Blue = 8 bits

Therefore:

$$\text{Bits/pixel} = 8 + 8 + 8 = 24 \text{ bits/pixel} = \boxed{24 \text{ bits/pixel}}$$

(c) Calculate the uncompressed image size in bytes and MB.

Total bits:

$$2,073,600 \times 24 = 49,766,400 \text{ bits} = 49,766,400 \text{ bits}$$

Convert bits to bytes:

$$\frac{49,766,400}{8} = 6,220,800 \text{ bytes} = \boxed{6,220,800 \text{ bytes}}$$

Using:

$$1 \text{ MB} = 1,048,576 \text{ bytes}$$

$$\text{MB} = \frac{6,220,800}{1,048,576} \approx 5.93 \text{ MB} = \boxed{5.93 \text{ MB}}$$

Therefore, the uncompressed image size is approximately **5.93 MB**.

(d) What happens if the image is converted to an 8-bit grayscale image?

A grayscale image uses only one channel.

Therefore:

Bits/pixel=8Bits/pixel=8

Storage required:

$2,073,600 \times 82,073,600 \times 8 = 16,588,800 \text{ bits} = 16,588,800 \text{ bits}$

Convert to bytes:

$16,588,800 \div 8 = 2,073,600 \text{ bytes}$ $\frac{16,588,800}{8} = 2,073,600 \text{ bytes}$

Therefore:

$2,073,600 \text{ bytes} \approx 1.98 \text{ MB}$ $\boxed{2,073,600 \text{ bytes} \approx 1.98 \text{ MB}}$

Comparison

Image	Bits/Pixel	Approx. Size
RGB	24	5.93 MB
Grayscale	8	1.98 MB

The grayscale image requires approximately **one-third of the uncompressed RGB storage**.

However, color information is lost. Brightness/detail information is retained as grayscale intensity.

(e) Recommend a suitable file format for high-quality archival storage.

TIFF is a suitable choice for high-quality archival storage because it supports high-quality image data and can use lossless compression.

For professional medical imaging, **DICOM** would be preferred when the image is part of a medical-imaging system because it is designed for medical image data and associated metadata.

Q3. A computer-vision application has to detect objects that may touch horizontally, vertically or diagonally in a binary image. Using a suitable sample matrix, analyze

4-neighborhood, 8-neighborhood, 4-adjacency, 8-adjacency and connectivity.

Consider the following binary image:

$$A = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

Consider the center pixel:

$$p = (2,2)$$

4-Neighborhood

The 4-neighbors are:

$$(1,2), (2,1), (2,3), (3,2)$$

All four contain 0.

Therefore, the center pixel has **no 4-connected neighboring 1-pixel**.

8-Neighborhood

The four diagonal pixels are also included:

$$(1,1), (1,3), (3,1), (3,3)$$

All four contain 1.

Therefore, the center pixel has four diagonal 8-neighbors.

4-Adjacency

Two pixels are 4-adjacent when they are horizontal or vertically adjacent and belong to the specified pixel set.

The center pixel is not 4-adjacent to the corner pixels.

8-Adjacency

Diagonal pixels are also considered adjacent.

Therefore, the center pixel is 8-adjacent to all four corner pixels.

Connectivity

Under **4-connectivity**, the five 1-pixels form five separate objects.

Under **8-connectivity**, all five 1-pixels are connected.

Therefore:

4-connectivity=5 objects
8-connectivity=1 object

Distance Measures

For two pixels $p(x_1, y_1)$ and $q(x_2, y_2)$:

Euclidean

$$D_E = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

City-Block

$$D_4 = |x_1 - x_2| + |y_1 - y_2|$$

Chessboard

$$D_8 = \max(|x_1 - x_2|, |y_1 - y_2|)$$

Conclusion

The choice of connectivity significantly affects object detection. **4-connectivity** recognizes only horizontal and vertical connections, whereas **8-connectivity** also recognizes diagonal connections.

Q4. A color-image application receives color images from a camera but needs different representations for display, printing, color-based segmentation and video transmission. Explain which among RGB, CMY(K), HSI and YCbCr would be suitable for each requirement.

1. Display – RGB

RGB is most suitable for display devices.

It contains:

- Red
- Green
- Blue

Computer monitors, TVs and mobile displays generally use additive RGB representation.

Why?

The display generates colors by combining red, green and blue light.

2. Printing – CMY(K)

CMY(K) is suitable for printing.

Components:

- Cyan
- Magenta
- Yellow
- Black

Why?

Printing works by absorbing light using inks, so a subtractive color model is appropriate.

3. Color-Based Segmentation – HSI

HSI is suitable for color-based segmentation.

Components:

- Hue
- Saturation
- Intensity

Why?

Hue and saturation describe color separately from brightness.

For example, an application can select pixels having a particular hue to identify a colored object.

4. Video Transmission – YCbCr

YCbCr is suitable for video transmission.

Components:

- Y = luminance
- Cb = chrominance

- Cr = chrominance

Why?

Human vision is more sensitive to luminance detail than chrominance detail. Therefore, chrominance information can be represented more efficiently for compression and transmission.

Summary

Requirement	Suitable Model	Reason
Display	RGB	Matches display technology
Printing	CMY(K)	Subtractive color model
Color segmentation	HSI	Separates color from intensity
Video transmission	YCbCr	Efficient luminance/chrominance representation

Q5. A satellite image must be geometrically analyzed before identifying roads and regions. Explain how two-dimensional mathematical tools and image geometry can be used to represent pixel coordinates, calculate distances, perform translation, scaling and rotation, and analyze relationships between pixels. Illustrate the answer with suitable mathematical expressions/examples.

Satellite images contain large amounts of spatial information. Mathematical operations are required to analyze roads, buildings, regions and other objects.

1. Pixel Coordinates

Each pixel can be represented as:

$$p=(x,y)p=(x,y)$$

For example:

$$p=(20,30)p=(20,30)$$

means that the pixel is located at coordinate (20,30)(20,30).

2. Distance Between Pixels

Suppose:

$$p=(2,3) \quad q=(2,3)$$

and:

$$q=(6,6) \quad q=(6,6)$$

Euclidean Distance

$$D_E = \sqrt{(6-2)^2 + (6-3)^2} = \sqrt{16+9} = 5 = \boxed{5}$$

City-Block Distance

$$D_4 = |6-2| + |6-3| = 4+3 = 7 = \boxed{7}$$

Chessboard Distance

$$D_8 = \max(4, 3) = 4 = \boxed{4}$$

3. Translation

Translation moves a pixel or object from one location to another.

If:

$$p=(x,y) \quad p=(x,y)$$

and translation is:

$$(t_x, t_y) \quad (t_x, t_y)$$

then:

$$x' = x + t_x \quad y' = y + t_y$$

Example

For:

$$p=(10,20) \quad p=(10,20)$$

and:

$$(t_x, t_y) = (5, 8) \quad (t_x, t_y) = (5, 8)$$

we get:

$$p' = (15, 28) \quad p' = (15, 28)$$

4. Scaling

Scaling changes the size of an image.

$$x' = s_x x \quad y' = s_y y$$

If:

$$p = (10, 20) \quad p = (10, 20)$$

and:

$$s_x = s_y = 2 \quad s_x = s_y = 2$$

then:

$$p' = (20, 40) \quad p' = (20, 40)$$

The object becomes twice as large.

5. Rotation

Rotation changes the orientation of an image.

For rotation by angle θ :

$$x' = x \cos \theta - y \sin \theta \quad y' = x \sin \theta + y \cos \theta$$

For 90° :

$$\cos 90^\circ = 0 \quad \sin 90^\circ = 1$$

Therefore:

$$x' = -y \quad y' = x$$

6. Homogeneous Coordinates

Geometric transformations can also be represented using matrices.

A point can be represented as:

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Translation can be represented as:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

This is useful for combining multiple geometric transformations.

7. Pixel Relationships

Satellite-image analysis uses:

4-Neighborhood

Horizontal and vertical neighbors.

8-Neighborhood

Horizontal, vertical and diagonal neighbors.

These relationships help identify connected roads and regions.

8. Application to Road Detection

For example, roads may appear as connected regions with similar intensity or color.

The system can:

1. Convert the satellite image into a suitable representation.
2. Enhance contrast.
3. Segment road-like regions.
4. Analyze connected pixels.
5. Calculate distances and boundaries.
6. Apply geometric transformations.

7. Identify roads and regions.

Conclusion

Two-dimensional mathematical tools provide the foundation for geometric analysis of satellite images. Coordinates, distance measures, connectivity, translation, scaling and rotation allow roads, buildings and geographical regions to be accurately represented and analyzed.